LAB 2: DATABASE DESIGN AND NORMALIZATION

VUA-SEN307 | Week 4

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1. Unnormalized Table

The original registration sheet contains multiple course values in some cells.

matric_no	student_name	student_dept	course_codes	course_titles	lecturers	lecturer_depts	session
20231234	Ada Obi	Software Engineering	SEN307, SEN301	Database Systems, Software Architecture	Dr A. Okafor, Dr M. Bello	Software Eng., Software Eng.	2026/2027
20231235	John Musa	Computer Science	SEN307, COS301	Database Systems, Operating Systems	Dr A. Okafor, Mrs F. Eze	Software Eng., Computer Sci.	2026/2027
20231236	Grace Okon	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Eng.	2026/2027

2. First Normal Form (1NF)

Primary key: (matric_no, course_code)

matric_no	student_name	student_dept	course_code	course_title	lecturer	lecturer_dept	session
20231234	Ada Obi	Software Engineering	SEN307	Database Systems	Dr A. Okafor	Software Eng.	2026/2027
20231234	Ada Obi	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Eng.	2026/2027
20231235	John Musa	Computer Science	SEN307	Database Systems	Dr A. Okafor	Software Eng.	2026/2027
20231235	John Musa	Computer Science	COS301	Operating Systems	Mrs F. Eze	Computer Sci.	2026/2027
20231236	Grace Okon	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Eng.	2026/2027

What I fixed: I split the multi-valued cells so that each student–course pair has its own row and every cell holds one value only.

3. Second Normal Form (2NF)

Student — matric_no (PK), student_name, student_dept

matric_no	student_name	student_dept
20231234	Ada Obi	Software Engineering
20231235	John Musa	Computer Science
20231236	Grace Okon	Software Engineering

Course — course_code (PK), course_title, lecturer, lecturer_dept

course_code	course_title	lecturer	lecturer_dept
SEN307	Database Systems	Dr A. Okafor	Software Eng.
SEN301	Software Architecture	Dr M. Bello	Software Eng.
COS301	Operating Systems	Mrs F. Eze	Computer Sci.

Enrolment — matric_no (PK/FK), course_code (PK/FK), session

matric_no	course_code	session
20231234	SEN307	2026/2027
20231234	SEN301	2026/2027
20231235	SEN307	2026/2027
20231235	COS301	2026/2027
20231236	SEN301	2026/2027

What I fixed: student_name and student_dept depend only on matric_no, and course_title depends only on course_code, so I moved them into their own Student and Course tables.

4. Third Normal Form (3NF)

Department — dept_id (PK), dept_name

dept_id	dept_name
1	Software Engineering
2	Computer Science

Student — student_id (PK), matric_no, full_name, dept_id (FK)

student_id	matric_no	full_name	dept_id
1	20231234	Ada Obi	1
2	20231235	John Musa	2
3	20231236	Grace Okon	1

Lecturer — lecturer_id (PK), full_name, dept_id (FK)

lecturer_id	full_name	dept_id
1	Dr A. Okafor	1
2	Dr M. Bello	1
3	Mrs F. Eze	2

Course — course_id (PK), course_code, course_title

course_id	course_code	course_title
1	SEN307	Database Systems
2	SEN301	Software Architecture
3	COS301	Operating Systems

Teaching — teaching_id (PK), course_id (FK), lecturer_id (FK), session

teaching_id	course_id	lecturer_id	session
1	1	1	2026/2027
2	2	2	2026/2027
3	3	3	2026/2027

Enrolment — enrolment_id (PK), student_id (FK), course_id (FK), session

enrolment_id	student_id	course_id	session
1	1	1	2026/2027
2	1	2	2026/2027
3	2	1	2026/2027
4	2	3	2026/2027
5	3	2	2026/2027

What I fixed: student_dept and lecturer_dept describe a department rather than the student or lecturer, so I moved department details into a Department table linked by a foreign key. Which lecturer teaches which course in which session is stored in the Teaching table. If a department is renamed, it is changed in one place only.

5. Final Schema

Table	Attributes and relationships
Department	dept_id (PK), dept_name
Student	student_id (PK), matric_no, full_name, dept_id (FK) → Department
Lecturer	lecturer_id (PK), full_name, dept_id (FK) → Department
Course	course_id (PK), course_code, course_title
Teaching	teaching_id (PK), course_id (FK) → Course, lecturer_id (FK) → Lecturer, session
Enrolment	enrolment_id (PK), student_id (FK) → Student, course_id (FK) → Course, session

Key: PK = Primary Key; FK = Foreign Key.